

Creating a CloudTrail Trail and EventBridge Alert for Console Sign-Ins

Christopher Samugana

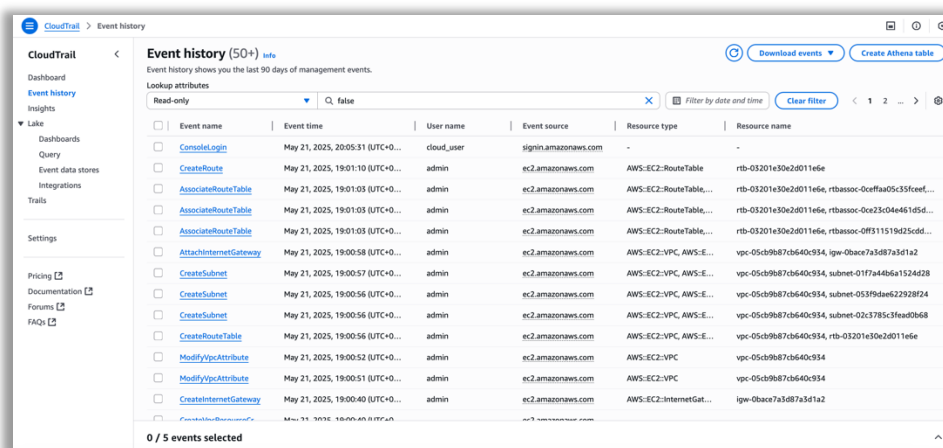
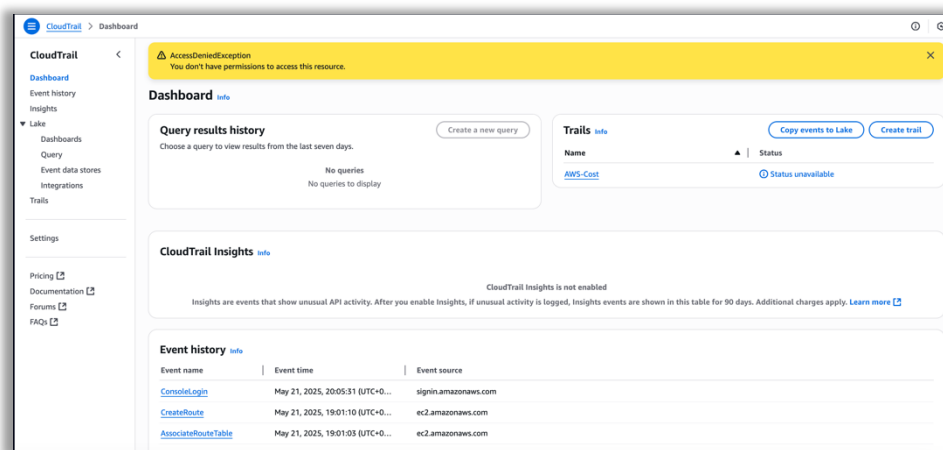
Introduction

In this project, I set up monitoring and alerting for AWS Console Sign-In activity by creating a CloudTrail Trail, configuring an SNS Topic and Subscription, and establishing an EventBridge Rule to notify on login events.

1. Created a Multi-Region CloudTrail Trail

I began by logging into the AWS Management Console using the provided credentials and ensured I was operating in the us-east-1 region.

I navigated to CloudTrail, explored the Event history to view the past 90 days of logged events, and noted the structure of sign-in events.



From the Trails section, I selected Create trail. I named the trail OurTrail, chose to create a new S3 bucket for log storage, and verified that the bucket name populated correctly.

The screenshot shows the 'Choose trail attributes' step in the AWS CloudTrail console. The left sidebar indicates the progress: Step 1 (Choose trail attributes), Step 2 (Choose log events), and Step 3 (Review and create). The main content area is titled 'Choose trail attributes' and includes the following sections:

- General details:** A trail created in the console is a multi-region trail. [Learn more](#)
- Trail name:** Enter a display name for your trail. The name 'OurTrail' is entered in the text field. Below the field, it states: '3-128 characters. Only letters, numbers, periods, underscores, and dashes are allowed.'
- Storage location:** Two options are available:
 - ☒ **Create new S3 bucket:** Create a bucket to store logs for the trail.
 - ☐ **Use existing S3 bucket:** Choose an existing bucket to store logs for this trail.
- Trail log bucket and folder:** Enter a new S3 bucket name and folder (prefix) to store your logs. Bucket names must be globally unique. The text 'aws-cloudtrail-logs-417165250702-SRD43810' is entered in the text field. Below the field, it states: 'Logs will be stored in aws-cloudtrail-logs-417165250702-SRD43810/AmazonLogs/417165250702'.
- Log file SSE-KMS encryption:** ☒ **Enabled**. [Info](#)
- Customer managed AWS KMS key:** ☒ **New**, ☐ **Existing**

The screenshot shows the Amazon S3 console for the bucket '417165250702/'. The left sidebar shows the navigation menu with 'Amazon S3' selected. The main content area is titled '417165250702/' and includes the following sections:

- Objects (2):** A table listing the objects in the bucket. The table has columns for Name, Type, Last modified, Size, and Storage class. The objects listed are 'CloudTrail-Digest/' and 'CloudTrail/'.
- Properties:** A section for bucket properties, currently empty.

I enabled SSE-KMS encryption for the log files, selected to create a new KMS key, and entered the alias CloudTrailKey. I ensured log file validation was turned on. After choosing Management events as the event type and selecting both Read and Write API activity, I completed the setup by creating the trail.

The screenshot shows the 'Choose log events' step in the AWS CloudTrail console. The left sidebar indicates the progress: Step 1 (Choose trail attributes), Step 2 (Choose log events), and Step 3 (Review and create). The main content area is titled 'Choose log events' and includes the following sections:

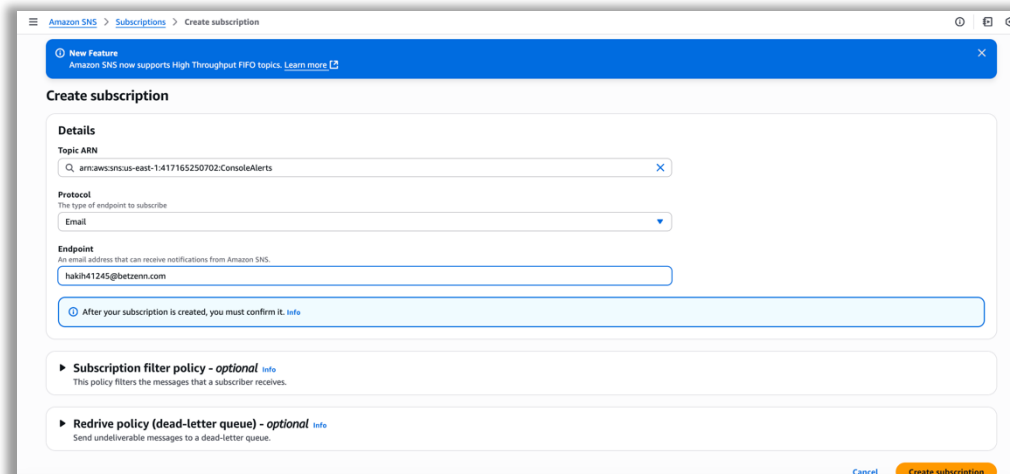
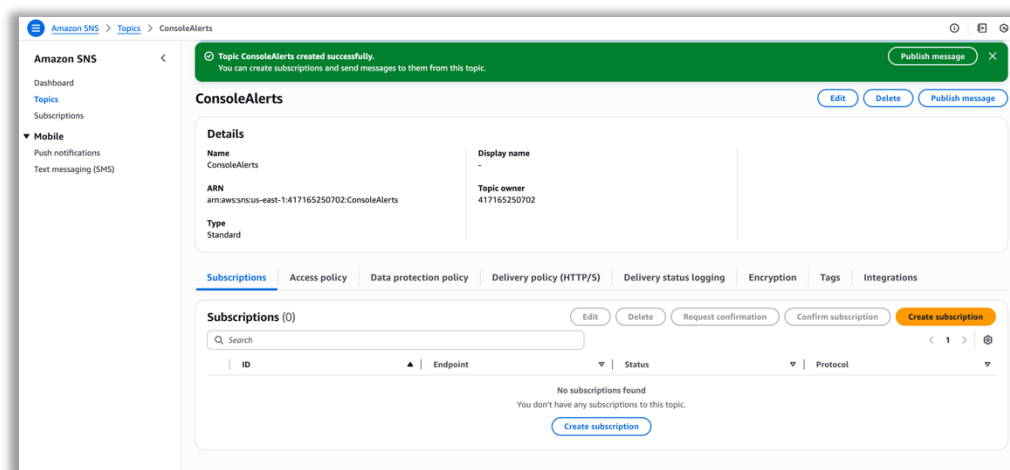
- Events:** Record API activity for individual resources, or for all current and future resources in AWS account. [Additional charges apply](#)
- Event type:** Choose the type of events that you want to log.
 - ☒ **Management events:** Capture management operations performed on your AWS resources.
 - ☐ **Data events:** Log the resource operations performed on or within a resource.
 - ☐ **Insights events:** Identify unusual activity, errors, or user behavior in your account.
 - ☐ **Network activity events:** Network activity events provide information about resource operations performed on a resource within a virtual private cloud endpoint.
- Management events:** [Info](#)
Management events show information about management operations performed on resources in your AWS account.
 - ☒ **No additional charges apply to log management events on this trail because this is your first copy of management events.**
- API activity:** Choose the activities you want to log.
 - ☒ **Read**, ☒ **Write**
 - ☐ **Exclude AWS KMS events**
 - ☐ **Exclude Amazon RDS Data API events**

B. Created a CloudTrail-specific KMS Key

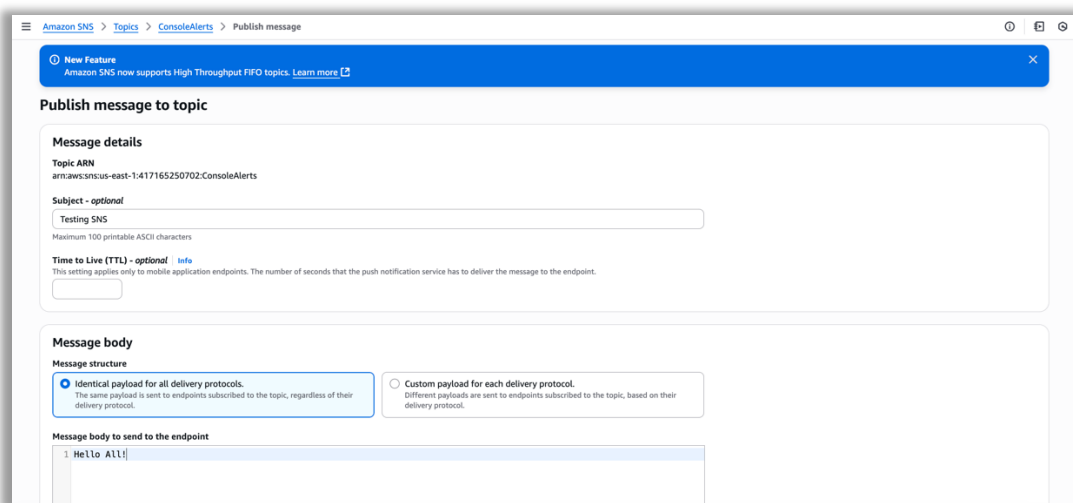
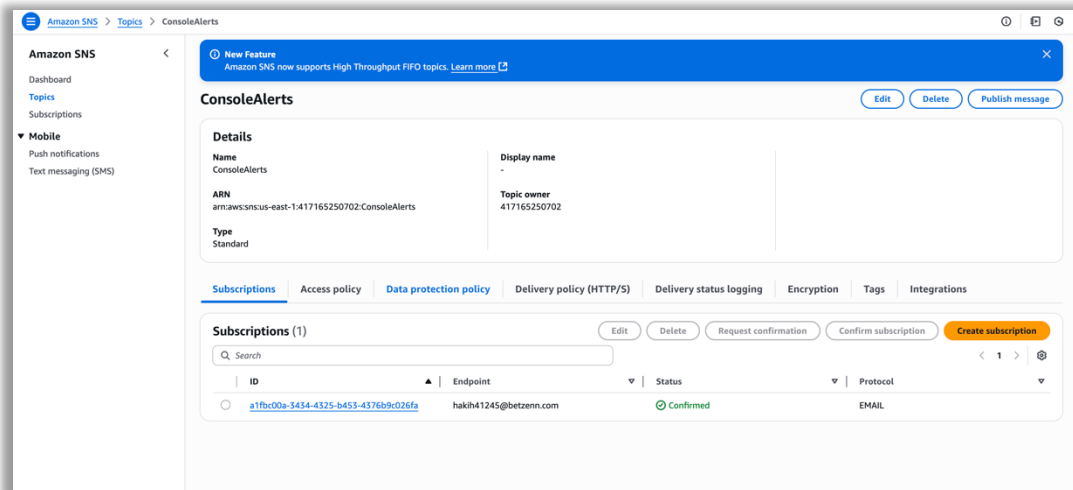
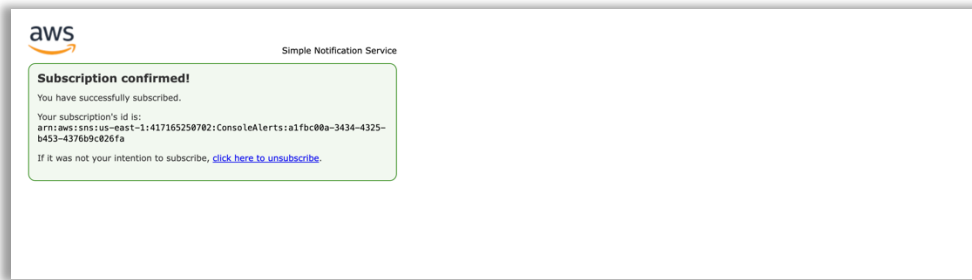
During the trail creation process, I opted to use SSE-KMS encryption and created a new KMS key labelled CloudTrailKey to ensure secure log storage. I enabled log file validation to help detect any modifications or deletions after delivery.

C. Created an SNS Topic and Subscription

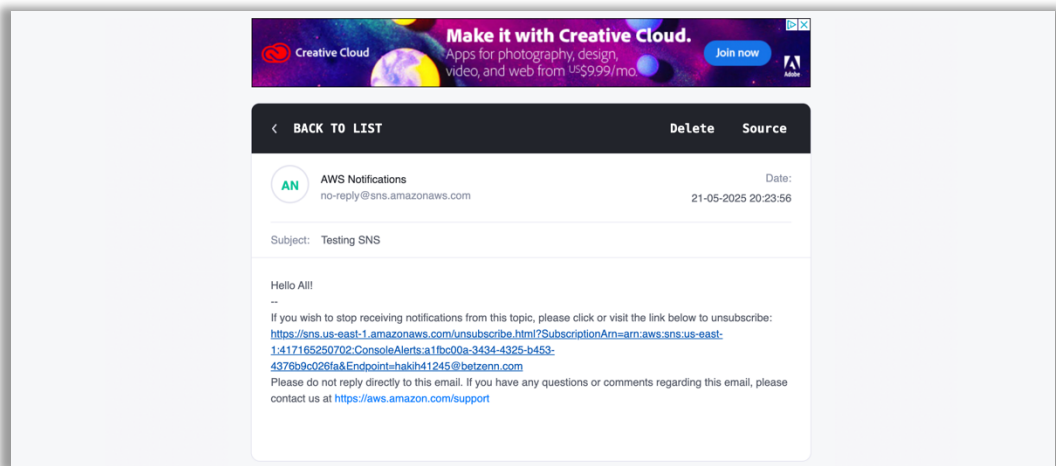
I navigated to Amazon SNS, selected Topics, and created a new Standard topic named ConsoleAlerts. Next, I created a subscription under this topic using Email as the protocol and provided a temporary email address for testing.



After submitting the subscription, I confirmed it through the verification link received in the inbox. I validated the confirmation by checking the subscription status in the SNS console, which showed as Confirmed.

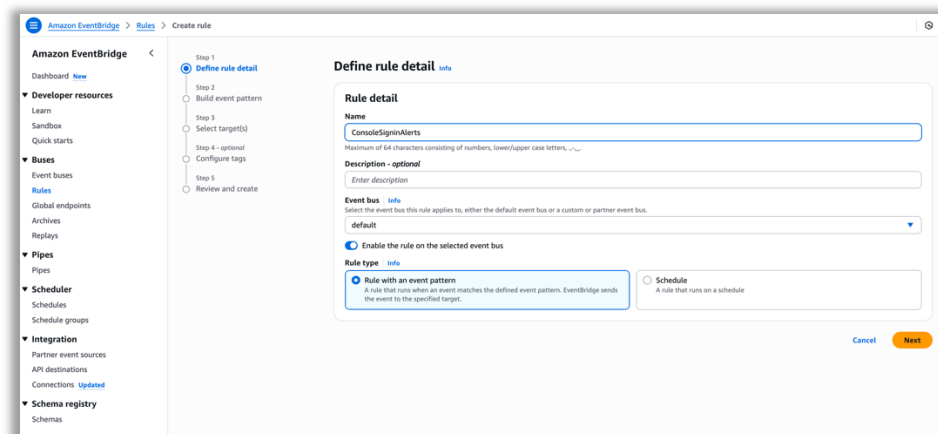


To test the subscription, I published a message to the topic using the Publish message feature, entered a sample subject and message body, and confirmed that the email was received successfully.

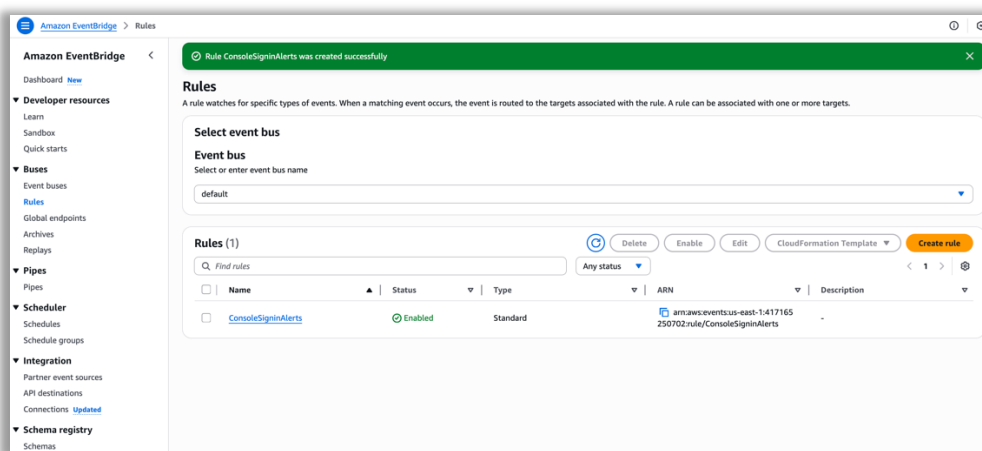
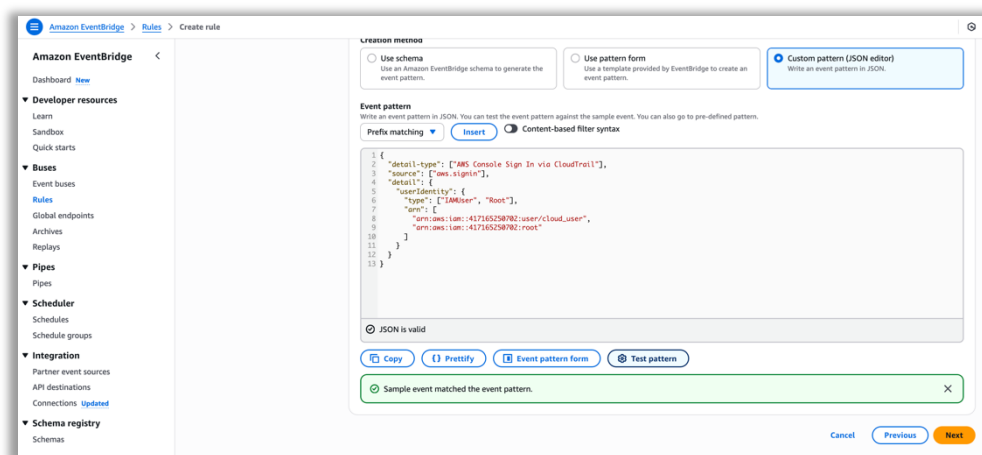


D. Set Up an Amazon EventBridge Rule

I proceeded to configure Amazon EventBridge to trigger alerts based on specific sign-in events. From the Rules section, I selected Create rule, named it ConsoleSigninAlerts, and enabled it on the default event bus.



For the event pattern, I used the Custom pattern (JSON editor) option and entered a JSON structure targeting AWS Console Sign In via CloudTrail events. I replaced all REPLACE_ME placeholders in the arn fields with my actual AWS account ID (680829786361) to filter events specific to my user (cloud_user) and the root user.



After validating the pattern using the Test pattern button and confirming a successful match, I selected SNS Topic as the target service and chose the previously created ConsoleAlerts topic.

I skipped to the review and finalized the rule creation.

E. Tested the Alert System

To verify the alert system, I signed out of the AWS Console and signed back in using the cloud_user account credentials provided by the lab. Shortly after, I received an email from the SNS topic containing the full JSON payload of the captured login event, confirming that the EventBridge rule and SNS integration worked as intended.

BACK TO LIST

DeleteSource

AN

AWS Notifications

no-reply@sns.amazonaws.com

Date:
21-05-2025 20:37:28

Subject: AWS Notification Message

[{"version":"0","id":"bf9109-80d0-2e59-b0e7-782478cd9fc","detail-type":"AWS Console Sign In via CloudTrail","source":"aws.signin","account":"417165250702","time":"2025-05-21T19:38:55Z","region":"us-east-1","resources":["detail":{"eventVersion":"1.09","userIdentity":{"type":"IAMUser","principalId":"AIDAWCII4SCHHVXNAYSLK","arn":"arn:aws:iam::417165250702:user/cloud_user","accountId":"417165250702","userName":"cloud_user"},"eventTime":"2025-05-21T19:38:55Z","eventSource":"signin.amazonaws.com","eventName":"ConsoleLogin","awsRegion":"global","sourceIPAddress":"94.65.97.240","userAgent":"Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/132.0.0.0 Safari/537.36","OPRV":"117.0.0.0","requestParameters":null,"responseElements":{"ConsoleLogin":{"Success":true,"additionalEventData":{"LoginTo":"https://us-east-1.console.aws.amazon.com/console/home?hashArgs=%5C23&isauthcode=true&oauthStart=1747856198235&om=us-east-1&state=hashArgFromTB_us-east-1_50-c165d78d13d31","MobileVersion":"No","MPAUsed":"No"},"eventId":"5dfcd8f-04b2-4e63-b51e-074c38dd8bbd","readOnly":false,"eventType":"AwsConsoleSignIn","managementEvent":true,"recipientAccountId":"417165250702","eventCategory":"Management","tlsDetails":{"tlsVersion":"TLSv1.3","cipherSuite":"TLS_AES_128_GCM_SHA256","clientProvidedHostHeader":"signin.amazonaws.com"}}]}]}]

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<https://sns.us-east-1.amazonaws.com/unsubscribe.htm?SubscriptionArn=arn:aws:sns:us-east-1:417165250702:ConsoleAlerts:11b00a-3434-4325-b453-4376b0c026a8¬ification-hash=41245@heizern.com>

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